

Prime Factorization Compiler

B10
MISTERS

Md. Miraz Morshed 202414065

Farhan Sadik Tahsin 202414072

Prime Factorization Compiler

INPUT

factor(360);

OUTPUT

$2^3 \times 3^2 \times 5^1$

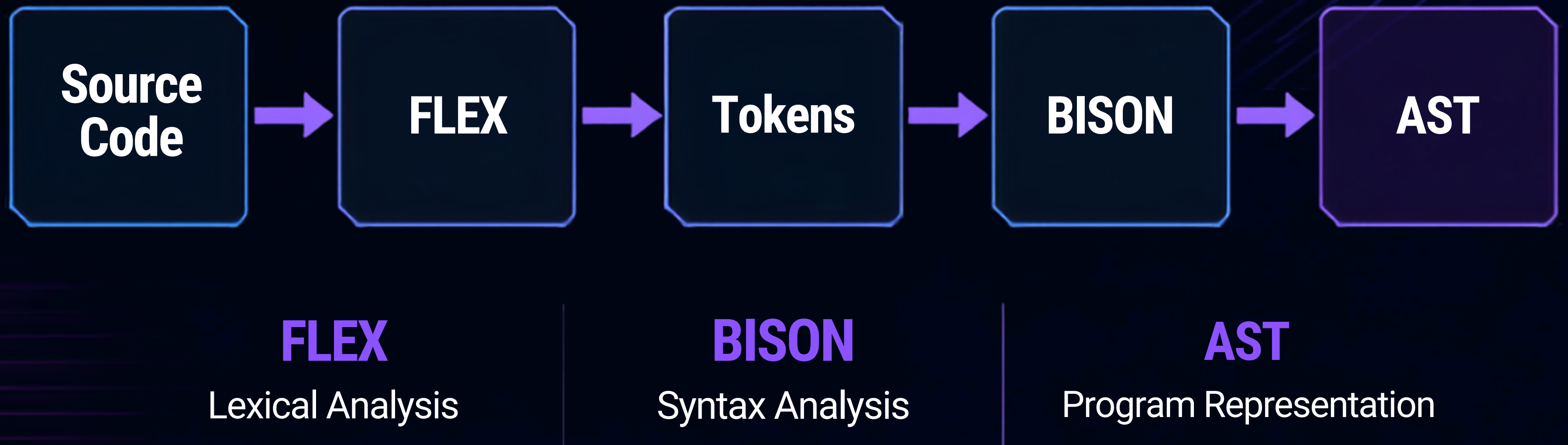
Building a compiler that transforms a high-level factorization statement into executable code.

SOURCE

COMPILER

OUTPUT

Phase 1 – Compiler Frontend

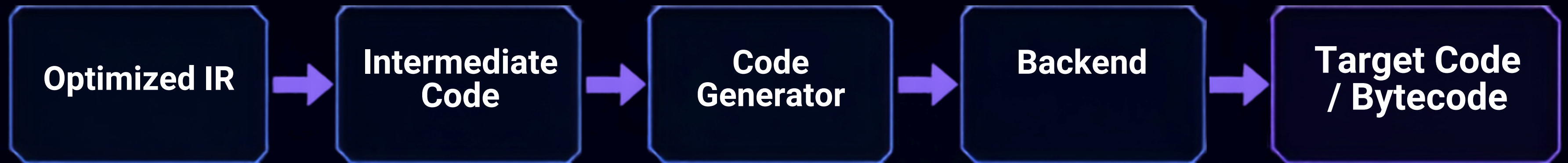


Phase 2 – Intermediate Representation



```
LOAD 360  
FACTOR  
EMIT 2,3  
EMIT 3,2  
EMIT 5,1
```

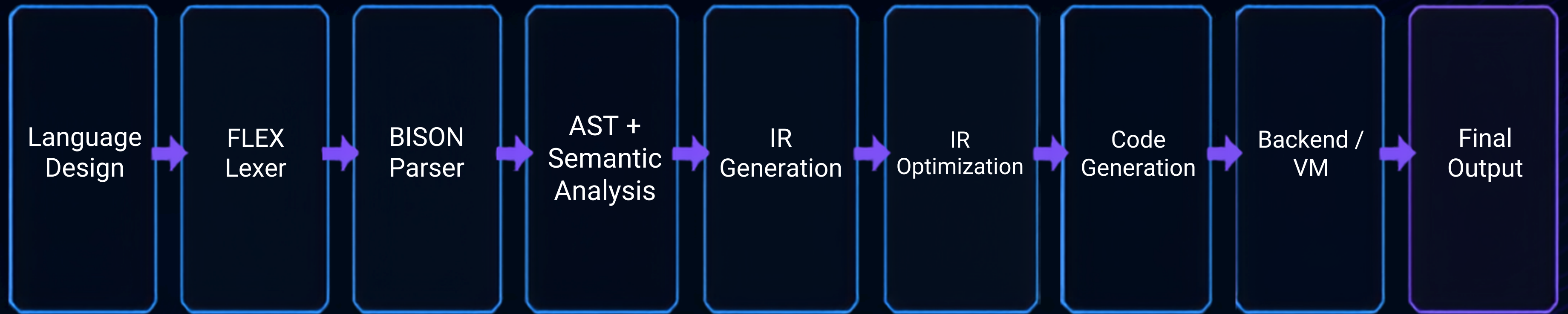

Phase 3 – Backend



CUSTOM BYTECODE + VIRTUAL MACHINE



Complete Development Roadmap



`factor(360);` → $2^3 \times 3^2 \times 5^1$

From Source Program to Executable Prime Factorization